P13: Building a Bee Monitoring Pipeline for Environmental Observation

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Context & Relevance

- Bees essential for food production and ecosystems.
- Bee populations are declining due to habitat loss and climate change.
- Manual monitoring not scalable at large scale.
- Continuous and automated large-scale monitoring.





*https://www.ualberta.ca/

*source:FAO (Food and Agriculture Organization of the United Nations)

Why Automated Bee Monitoring?

Traditional Monitoring
Manual process

Field Visit
On-site presence

→

Direct Observation
Labor intensive

→

Manual Counting
Error-prone

→

Statistics
Limited data



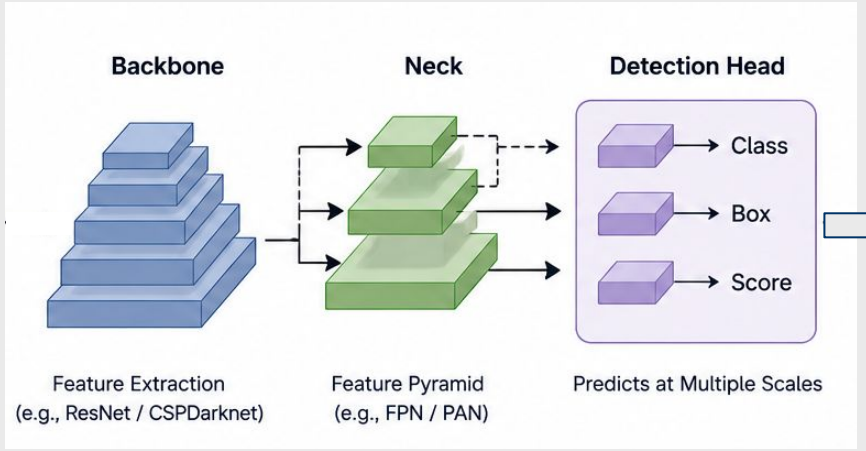


System architecture

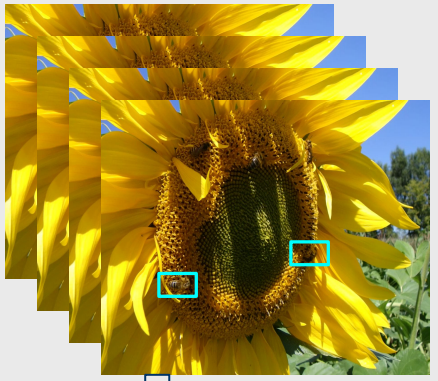
Video



Detection model



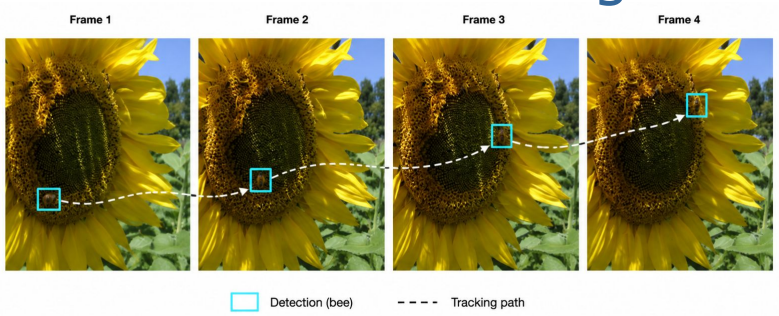
Detections per frame



Dashboard



Tracking



*Concept visualization — AI generated

Model Training Pipeline

Dataset



BuzzSet

Annotated flower images with target objects (bees) highlighted via bounding boxes.

→

Pretrained Models

YOLO

High-speed, real-time convolutional neural network detector.

RFDETR

Transformer-based model offering advanced localization accuracy.

→

Fine-Tuning

🎯

Fine Tuned Model

Customized bee detection model optimized for identifying small objects in dense backgrounds.

Dataset

BuzzSet

- Pre-existing dataset
- Single frame Dataset
- Total 8600 Annotated Images
- 5 object categories and different backgrounds

Class Distribution

1. Bee : 8566 (78.37%)
2. Bumble bee : 233 (2.13%)
3. Hoverfly : 1613 (14.76%)
4. Moth : 202 (1.85%)
5. Other : 316 (2.89%)



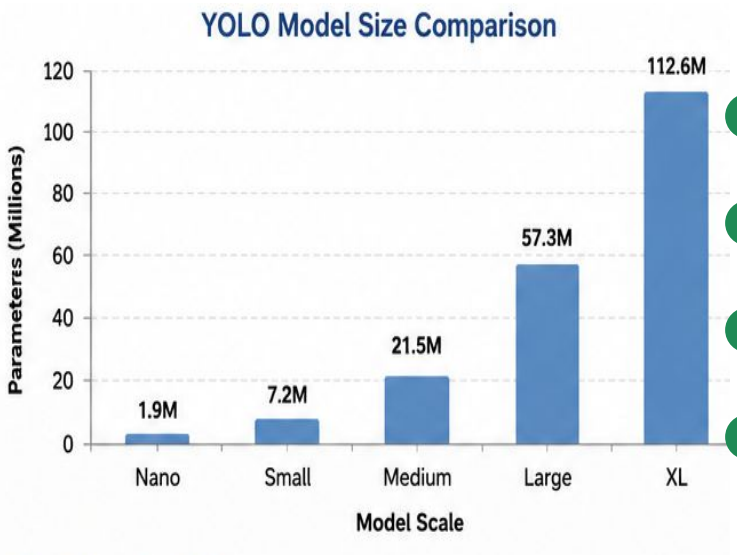



Choosing a Model

YOLO Family

RF-DETR (Transformer-based)

Scale: Nano → Small → Medium → Large → XL



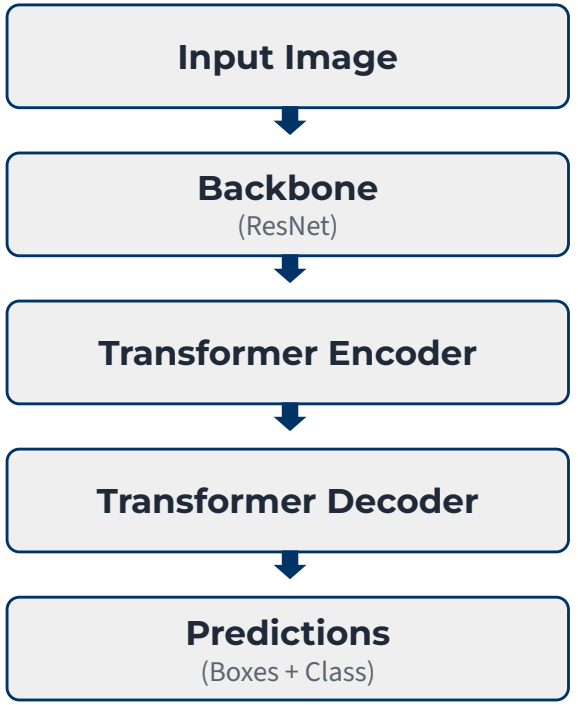
Key Characteristics

- CNN (Convolutional Neural Network) backbone
- Non-Maximum Suppression(NMS)
- Fast inference
- Great for small / real-time applications

Key Characteristics

- Residual network (ResNet) backbone + transformer encoder-decoder
- Set-based prediction, no NMS needed
- Stronger localization
- Higher computational cost, slower inference

Architecture Overview



End-to-end training • No NMS • Better localization

*https://rfdetr.roboflow.com/
*https://docs.ultralytics.com/

Model Evaluation

Model	Parameters	mAP@50–95*↑	mAP@50*↑	Precision*↑	Recall*↑
RF-DETR	128M	0.325	0.535	0.601	0.540
YOLO 26X	57M	0.322	0.484	0.512	0.452
YOLO 11L	25M	0.275	0.421	0.638	0.413
YOLO v8L	44M	0.223	0.411	0.701	0.400

**Our experimental evaluation on the pollinator dataset.*

Edge Computing Performance

Metric	YOLOv26X	YOLOv11L	YOLOv8L	RF-DETR
Params (M)	68	25	43	128
Size (MB)	118	50	87	140
FPS (Jetson Orin)*	15–25	30–45	20–35	10–20
Edge-deployable?	Maybe	Yes	Yes	No (server/GPU)

**Estimates from published Jetson Orin benchmarks (Ultralytics, RF-DETR literature) — not yet measured on our dataset/hardware.*

Selected Baselines for Evaluation

Head-to-head comparison of the two baseline models selected for pollinator detection

⚡ YOLOv11L

25.3M

Parameters

- ⏱️ Faster inference
- ✓ Good balance of speed and accuracy

VS

🎯 RF-DETR Large

128M

Parameters

- ▲ Higher accuracy & localization
- ✓ Better at small objects & cluttered scenes

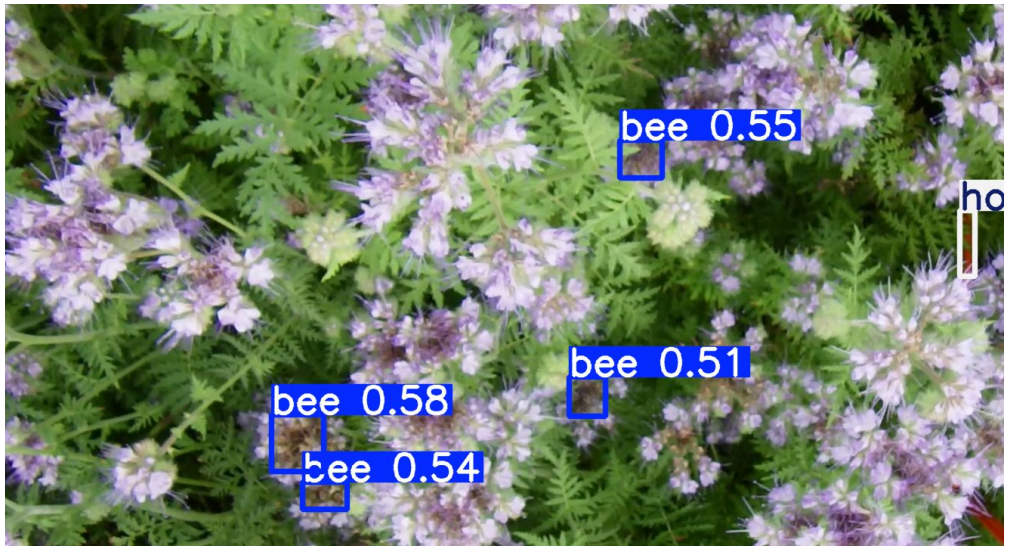
Key takeaway: YOLO models are efficient and fast, while RF-DETR provides stronger localization and performs better in dense and cluttered environments — making it a strong candidate for small pollinator detection.

RF-DETR Detections on unseen video

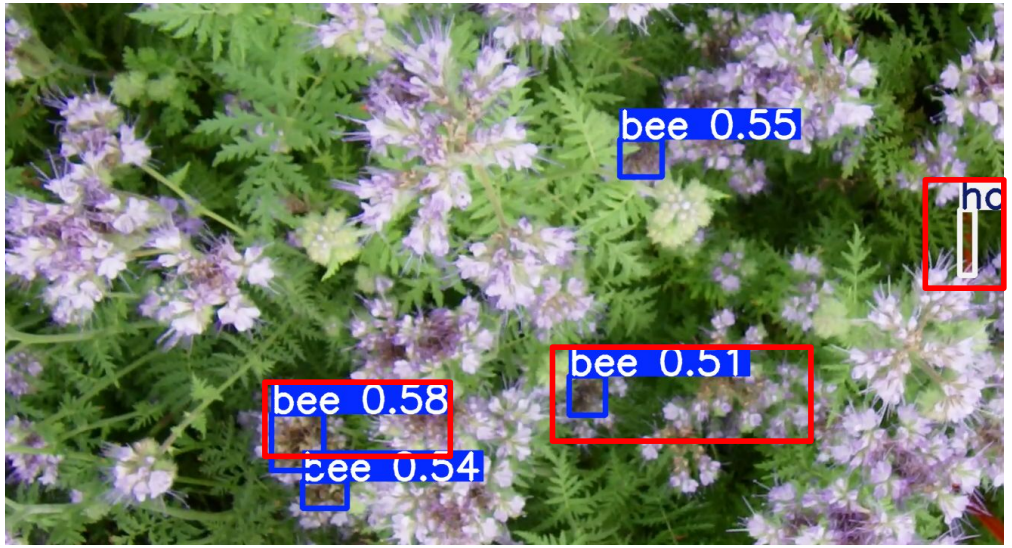


Major Problems

- False Positives
- False negatives
- Occlusion







Implementing SAHI

SAHI (Slicing Aided Hyper Inference) improves small-object detection by running inference on overlapping image slices.

1 Split into Slices

Divide the large image into overlapping patches.

2 Detect in Each Slice

Run the detector separately on each patch.

3 Map Back

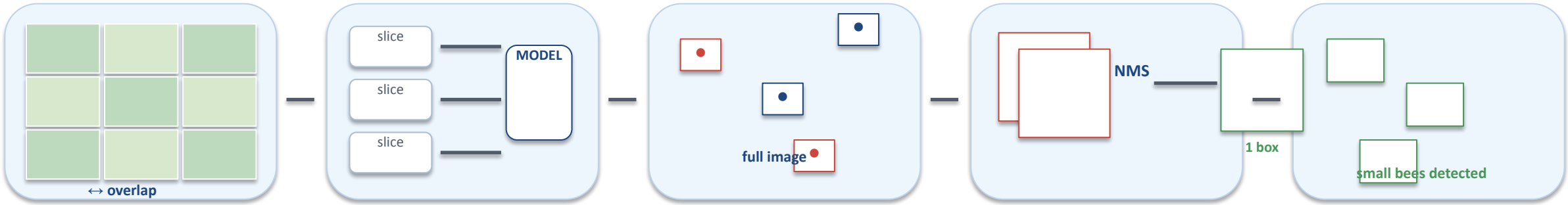
Convert slice coordinates to full-image coordinates.

4 Merge Detections

Use NMS to remove duplicate boxes.

5 Improved Detection

Better recall for small objects.



Why SAHI helps

- 🔍 Zooms in on small objects
- ⊞ Overlap reduces missed border objects
- ✓ Can improve recall / mAP
- 🐝 Useful for dense scenes with tiny bees

Key takeaway: SAHI lets the detector focus on smaller regions, making tiny pollinators easier to detect.

RF-DETR

Sliced



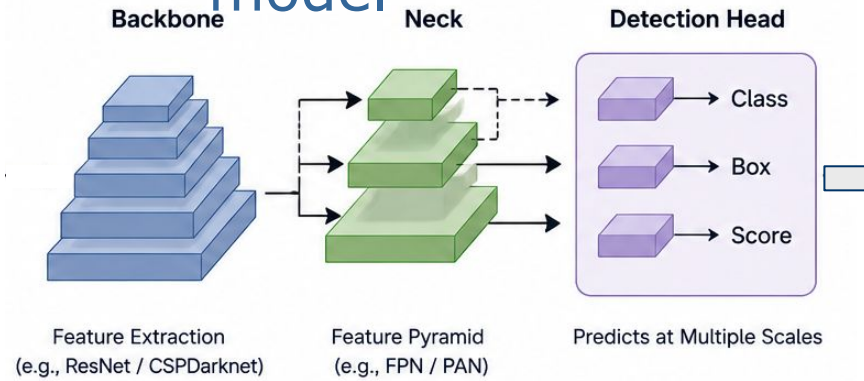
*image size: 512 × 512 pixels with 20% overlap

System architecture

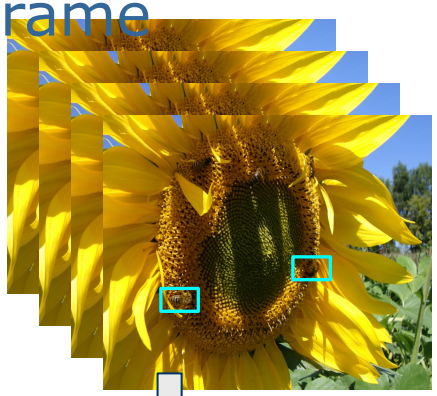
Video



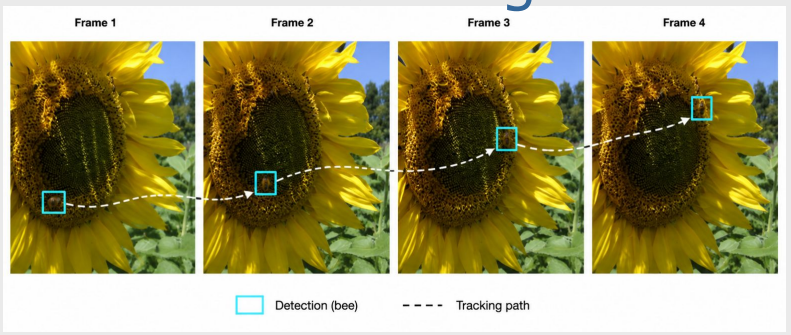
Detection model



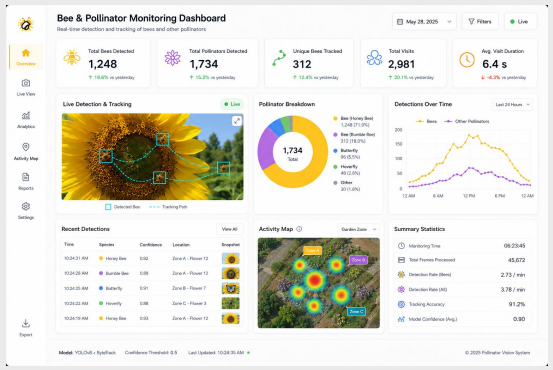
Detections per frame



Tracking



Dashboard



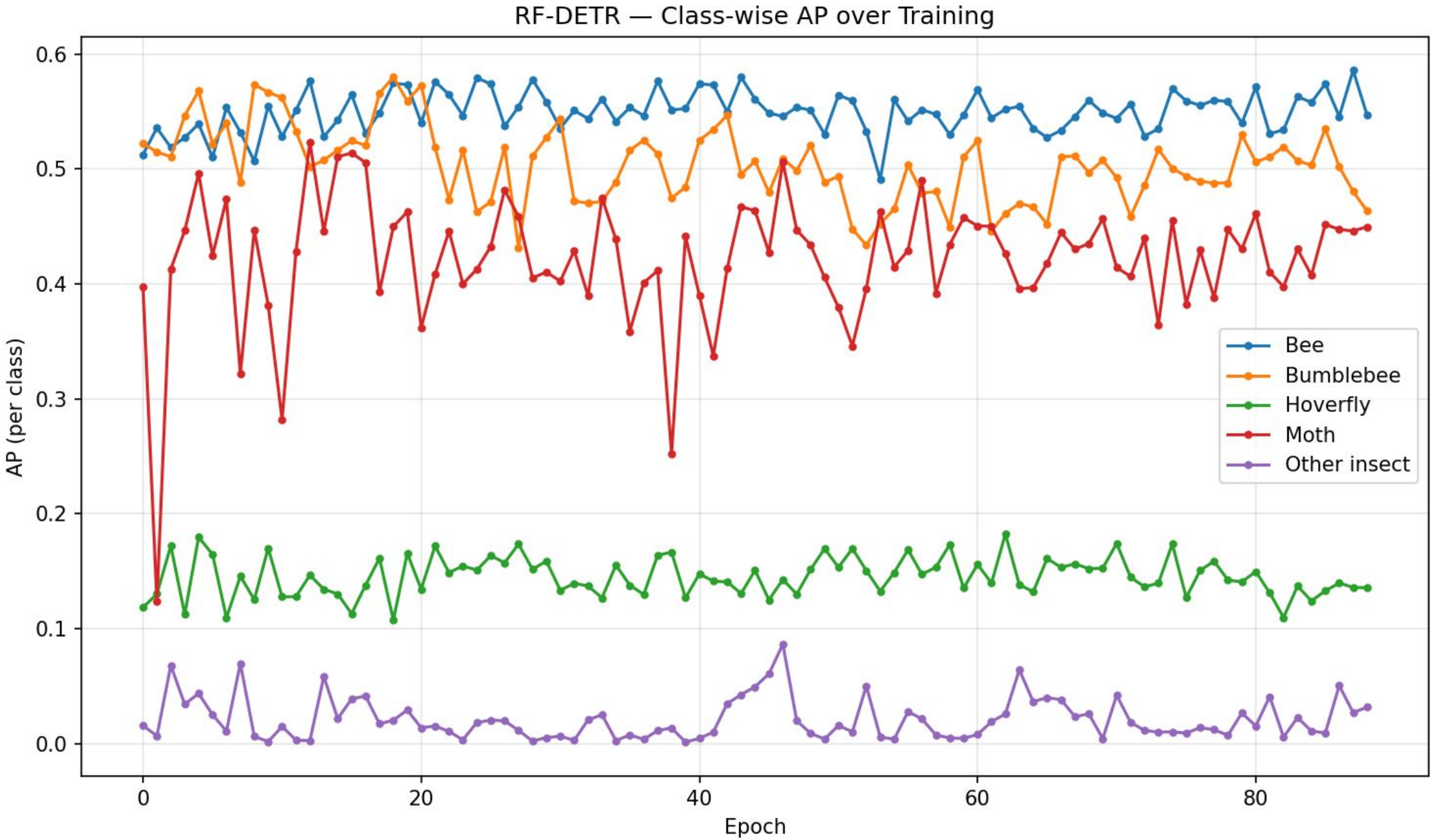
Part 2 Roadmap

- Expand dataset using more videos with difficult cases.
- Implement hard negative mining to Improve detector Model.
- Implement tracking.
- Refine analytics for better monitoring statistics.

Target Outcome

A more robust end-to-end monitoring system that works reliably on long, real-world videos.

Thank you


RF-DETR — Class-wise AP over Training
AP (per class)
Epoch
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0
20
40
60
80
Bee
Bumblebee
Hoverfly
Moth
Other insect

Part 2 roadmap

Expand dataset
More videos + difficult cases

Implement hard negative mining to Improve detector
Recall + localization

Implement tracking
Occlusion + ID stability

Refine analytics
Better monitoring statistics

Target outcome

A more robust end-to-end monitoring system that works reliably on long, real-world videos.

YOLOV11L



*img size: 1080 x 1080

YOLOV11L +



*img size: 1080 x 1080

Project Plan
(to be written but not to be presented)

Project Plan: Key Achievements Reached

- Goal: Describe the key achievements you have reached as a group. Also comment shortly, if plans have changed and why.

Tasks of Divyesh (s54dkana)	Tasks of Prince (s57pbamb)	Tasks of Aniket (s17alimb)	Tasks of Rohit (s09rrath)	Tasks of Person 5 (UniID)
- Trained the models and evaluate between each and every models. -Implement class balancing techniques - Experiment with different detection algorithms	- Literature review on bee detection and tracking methods - Compared YOLOv5, YOLOv7, and YOLOv8 models - Researched DeepSORT and ByteTrack tracking algorithms	- Studied environmental monitoring systems - Worked on evaluation metrics and research documentation -started creating the dataset from raw video - Trained YOLOv11n model	- Studied insect and pollinator monitoring datasets - Researched and compared detection models for bee monitoring - Trained YOLO11x Model.	List the performed tasks

Project Plan: Key Achievements Planned

- Goal: Describe the key achievements you have planned for the next period as a group.

Tasks of Divyesh (s54dkana)	Tasks of Prince (s57pbamb)	Tasks of Aniket (s17alimb)	Tasks of Rohit (s09rrath)	Tasks of Person 5 (UniID)
- Improve model accuracy using advanced methods - Implement class balancing techniques - Experiment with different detection algorithms - Optimize bee detection for small objects	- Apply data augmentation using Albumentations - Prepare train/validation/test datasets - Integrate DeepSORT and ByteTrack tracking - Improve video processing pipeline	- Research real-time deployment possibilities - Work on environmental analytics integration - Prepare final research documentation and reports	- Design and develop the Streamlit monitoring dashboard - Implement temporal bee activity charts - Write and compile the final project report	